DATA SCIENCE

Complete Course Document | Beginner to Intermediate

1. Course Overview

Data Science is a field that combines statistics, programming, mathematics, and analytical techniques to collect, process, analyze, and interpret data. It helps individuals and organizations discover patterns, answer questions, and make informed decisions.

2. Course Objectives

- Understand the fundamentals and applications of Data Science.
- Learn Python programming for data analysis.
- Collect, clean, and organize real-world datasets.
- Apply basic statistical methods and data visualization techniques.
- Understand introductory machine learning concepts.
- Complete practical exercises and a final project.

3. Course Curriculum

Module 1: Introduction to Data Science

- What is Data Science?
- Applications and use cases
- The Data Science lifecycle
- Types and sources of data
- Data collection methods

Module 2: Python for Data Science

- Python basics, variables, and data types
- Lists, tuples, sets, and dictionaries
- Conditions, loops, and functions
- File handling and working with data files

Module 3: Data Analysis

- Introduction to NumPy
- Pandas Series and DataFrames
- Data cleaning and preprocessing
- Handling missing and duplicate values
- Filtering, sorting, and grouping data

Module 4: Data Visualization

- Introduction to Matplotlib and Seaborn
- Bar charts and line charts
- Pie charts and histograms
- Scatter plots and distribution plots
- Interpreting charts and communicating findings

Module 5: Statistics for Data Science

- Mean, median, and mode
- Variance and standard deviation
- Basic probability
- Correlation and distributions
- Introduction to hypothesis testing

Module 6: Machine Learning Fundamentals

- Introduction to machine learning
- Supervised and unsupervised learning
- Linear regression and classification
- Decision trees
- Training, testing, and model evaluation

Module 7: Practical Projects

- Student performance analysis
- Sales data analysis
- Customer segmentation
- House price prediction

4. Tools and Technologies

Python • Jupyter Notebook • NumPy • Pandas • Matplotlib • Seaborn • Scikit-learn • SQL

5. Learning Outcomes

- Prepare and analyze structured datasets.
- Perform exploratory data analysis (EDA).
- Create visualizations to communicate patterns and trends.
- Apply introductory statistical and machine learning concepts.
- Build and explain small data-focused projects.

6. Career Opportunities

This course provides a foundation for roles such as Data Analyst, Junior Data Scientist, Business Intelligence Analyst, and further study toward Data Engineer or Machine Learning Engineer roles.

7. Course Completion

Learners should complete module exercises, quizzes, assignments, and a final practical project. A completion certificate may be issued according to the learning platform's course-completion rules.

Course Level: Beginner to Intermediate **Recommended Duration:** 8–12 Weeks
Prerequisites: Basic computer knowledge and logical thinking.